

Uziel Rodríguez Andújar

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Education

University of Puerto Rico at Mayagüez
Bachelor of Engineering in Mechanical Engineering

Expected Graduation Date: 06/2029
GPA: 3.9
Core Courses: Dynamics, Mechanics of Materials, Thermodynamics I

Technical Experience

Massachusetts Institute of Technology | 2.00C: Design for Complex Environmental Issues | January-May 2025
Undergraduate Teaching Fellow | The Lithium Challenge

- **Guided** and mentored students in **developing solutions** to complex, real-life problems.
- **Led** instructional **sessions** to **strengthen** students' research, and team communication skills.
- **Facilitated interdisciplinary collaboration**.

Bio-Inspired Perception and Robotics Laboratory (BPRL)

June -July 2026

Mechanical Engineering Intern

- **Designed** and analyzed structural components for a hybrid **VTOL quadcopter platform** integrating hover and cruise flight dynamics.
- **Performed** interference fit and structural simulation analysis on a custom snap-fit **PETG mounting bracket** in **SolidWorks**.
- **Developed** CAD assemblies incorporating **carbon fiber composite** structures and collaborated on component fabrication and assembly workflows.

University of Puerto Rico at Mayagüez | ROCS Carpool | Reserved Optimized Carpool Systems | March 2026-Present
Hardware Division

- **Contributed** to the design of a **smart** parking barrier, enabling operation through a mobile application.
- **Performed** Finite Element Analysis (**FEA**) to **evaluate** structural behavior, including stress distribution and deformation under applied loads.
- **Identified** discrepancies between simulation results and physical conditions, refining boundary conditions and modeling assumptions.
- **Supported electromechanical integration**, including electrical component layout and system interfacing.

University of Puerto Rico at Mayagüez | Co-Leader | Launch Initiatives for Technologies in Space | Sept 2025-Present
LIFTS | Payload Design and Manufacture Division

- **Designed** a payload that **integrates** all mission components within volume constraints.
- **Applied** aerospace and systems **engineering standards** to payload design, ensuring **NASA Balloon Program safety protocols**.
- **Conducted** risk assessments, **mechanical evaluations**, and coordinated **multidisciplinary** teams to implement **hazard controls** and **pre-flight verification procedures**.
- **Manufactured** the payload and conducted **test procedures** before launching.
- **Simulated** payload trajectories and **optimized** payload design for improved **efficiency** at target altitudes.

Massachusetts Institute of Technology | 2.00C: Design for Complex Environmental Issues |
Student Participant | Power Puzzle

- **Designed** and **prototyped** the "Gen Grip" to improve **usability** and **ergonomics** in power plant operations.
- Collaborated with a multidisciplinary team, both online and in person, to develop a **product** across all involved campuses.
- **Presented** and pitched the **prototype** to the clients, obtaining feedback for **refinement**.

Skills, Certifications & Achievements

- **Language:** English, Spanish
- **Technical Skills:** Solidworks, Microsoft 365, Python, RedCap, Soldering
- **Interpersonal Skills:** Leadership, Team Management, Teamwork, Critical Thinking, Problem Solving
- **Certification:** OSHA 29-CFR 1926 Construction Industry | ID No. 602559953
- **Recognition:** Dean's List Flag Bearer, UPR-Ponce; Psychology Award at National Math Fair